Chapter 5

Responsive features, the learner, and the population: efficient simulation of language contact dynamics

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A mathematical model of language contact from previous literature (Kauhanen 2022) predicts a critical threshold for contact-induced change: the proportion of second-language (L2) learners present in a population of speakers must exceed this threshold in order for simplification to occur. In this chapter, I investigate to what extent the model's predictions remain valid when key idealizing assumptions are dropped. Rather than working in the model's deterministic limit in which analytical results are available, the model is here studied using agent-based simulations, using realistically sized speech communities and locally structured interaction patterns. The key finding is that the analytical results obtained in the deterministic limit may very slightly overestimate the likelihood of contact-induced simplifications in language. Additionally, the interactions of several key model parameters are investigated, and the two specific case studies of Kauhanen (2022) are revisited using the agent-based methodology.

1 Introduction

Chapter **??** defined a *responsive feature* as any linguistic feature which is sensitive to the acquisition setting. In particular, L2-difficult features are responsive in this sense: by definition, such features pose a difficulty for, and only for, acquirers beyond a critical age. The Trudgill conjecture (Chapter **??**) predicts that such features are prone to being lost if enough acquirers beyond this critical age are found in the speech community. Writing on simplification of adjective inflection in Bergen Norwegian, Trudgill proposes:

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> My suggestion is that when the proportion of non-native becomes as close to 50% as this – the proportion of non-native speakers of Norwegian in Bergen was certainly very high (4,000 out of 9,000) – the number of face-to-face dialect-contact type interactions, and therefore potential instances of accommodation […], would have reached a threshold level at which some aspects of the non-native variety could transfer to the native (Trudgill 2011: pp. 57–58).

The present chapter undertakes a quantitative and computational recapitulation of this proposal, asking just how many post-critical-age acquirers must be present in the speech community for L2-difficult features to be lost. If there indeed exists a threshold of simplification due to the responsive nature of some or other feature, what is this threshold? Fifty percent L2 learners? Sixty percent? Or possibly a number whose precise value depends, in some way, on other factors involved in the contact situation? Moreover, how long must the contact situation continue for simplification to be permanent, so that the L2-difficult feature is not simply restored once L2 learners are removed from the population? Finally, how do sociodemographic factors such as the frequency of interactions between L2 learners and L1 speakers affect the likelihood of simplification?

The challenges involved in answering these questions ought to be clear from the outset. An empirical answer would require, at a minimum, a sample of language histories some of which culminate in loss of an L2-difficult feature, with others culminating in its retention, together with reliable estimates of the likely numbers of post-critical-age acquirers in each case, combined with information about further factors such as the degree of L2-difficulty exhibited by the feature in question, the duration for which L2 learning was in place, and the frequencies of different patterns of interaction between the different subpopulations. It is clear that empirical data are not available to this extent, at this resolution, in most cases of interest. As a consequence, a mixed methodology, combining empirical data, theory, and modelling, is called for.

The strategy adopted in the present chapter is to draw upon the predictive power of mechanistic modelling (Geritz & Kisdi 2012). It is first demonstrated how feature responsivity can be captured in a simple stochastic model of language acquisition. A mixed population is then introduced, one in which some learners acquire the language before the critical age, the remaining being post-critical-age acquirers; for ease of exposition, I will refer to the former as L1 learners, and the latter as L2 learners, in what follows.[1] As these learners interact, the

[1]It is important to bear in mind that, with this nomenclature, our L2 learners do not comprise

frequency of the L2-difficult feature may go up or down in the population; we are chiefly interested in the question of under what circumstances this frequency vanishes to zero, corresponding to full simplification.

The chapter builds upon work published in Kauhanen (2022). In that article, the above questions were answered in a particular limiting case of the model, known as the deterministic limit. This special case is characterized by the following idealizing assumptions: (i) there are infinitely many learners in each group (L1 and L2), (ii) learners pick interlocutors entirely at random, and (iii) all learners within a group behave the same. For completeness, the results of the earlier paper are restated and discussed below (in Section 3). However, the present chapter ventures beyond the deterministic limit, exploring a number of empirically relevant scenarios of the full stochastic model in computer simulations. A methodological innovation – efficient simulation of large populations through the use of a Beta approximation to describe the statistical equilibrium of language learning – is also introduced and discussed.

The results demonstrate that the stochastic simulations are broadly in line with the deterministic prediction, although the latter may slightly overestimate the likelihood of contact-induced simplification due to the fundamental well-mixing assumption of the deterministic-limit model. Simulations involving systematic sweeps across the model's parameter space uncover further, more fine-grained results, such as the (prima facie counterintuitive) fact that simplification is supported by population structures in which interactions between the L1 and L2 learner subpopulations are less frequent than interactions within subpopulations. Finally, two detailed simulation case studies, one on the expression of subjects in Afro-Peruvian Spanish, the other on the loss of verbal morphology in Afrikaans, support the claims that were made, based on the deterministic limit, in Kauhanen (2022).

2 Responsive features in language acquisition

Throughout this chapter, it is assumed that language learners need to decide between two grammars, one (and only one) of which carries a responsive feature.[2]

individuals who exhibit childhood bilingualism – such individuals learn an L2 but learn it before the critical age and hence the responsive features do not respond in their case. Moreover, empirical reality accommodates for more nuance than can be captured by a strictly binary cutoff between pre-critical age and post-critical age. See Chapter ?? for more discussion of these points.

[2]These assumptions are made in order to keep the model simple, without losing the essential intuition of competition between a responsive grammar and a non-responsive one. It would

I will denote the L2-difficult grammar with G_1 and its competitor with G_2. The grammars accept languages L_1 and L_2, respectively, understood as sets of well-formed strings, constructed out of a common alphabet. In general, L_1 and L_2 may overlap; cf. Figure 1. The symbol a_1 will be used to denote the probability of someone employing G_1 producing a string belonging to the set $L_1 \setminus L_2$ (i.e. a string that G_2 does not accept), and similarly a_2 will be used for the probability of someone employing G_2 producing a string in $L_2 \setminus L_1$. These quantities will be referred to as the *advantages* of the two grammars in what follows. They are also assumed to reflect the weak generative capacities of the two grammars (cf. Yang 2000) and may thus be assumed to be constant within any given competition situation.

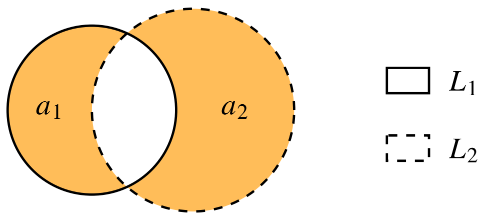

Figure 1: Two grammars, G_1 and G_2, generate languages L_1 and L_2, represented here as intersecting sets. The shaded areas are the sets $L_1 \setminus L_2$ and $L_2 \setminus L_1$; their sizes represent the probabilities that strings from these sets are produced, a_1 and a_2 (referred to as the advantages of G_1 and G_2, respectively).

Each language user is assumed to embody a pair of *weights* for the two grammars; formally, these weights constitute a categorical probability distribution over the grammars. The weight of G_1 will be denoted P and the weight of G_2 with $Q = 1 - P$; these are random variables whose values change in response to the linguistic input received by the language user or learner. These changes could in principle be implemented by any number of imaginable mechanisms, and it is one of the goals of language acquisition and learnability research to disentangle probable from improbable mechanisms of learning. In order to obtain a reasonably tractable mathematical model, I follow the variational learning framework of Yang (2002), itself founded upon the pioneering work of Bush & Mosteller (1955) on general statistical mechanisms of reinforcement.

be relatively straightforward to generalize the model for multiple competing grammars, each of which may involve some (potentially non-zero) amount of L2-difficulty. This and other potential avenues for further research are discussed at more length in Section 8.

Let us first outline how the learning algorithm works for an acquirer under the critical age. Upon receiving an input sequence $s \in L_1 \cup L_2$, the learner chooses one of the two grammars to use: G_1 with probability P, and G_2 with probability Q. The learner then attempts to parse the input sequence. If the chosen grammar accepts the input, this grammar's weight is increased and the weight of its competitor is decreased. If the grammar does not accept the input, the reverse operation is carried out, decreasing this grammar's weight and increasing the competitor's weight.

Concretely, the changes made to the weights follow the linear reward–penalty scheme of Bush & Mosteller (1955). The mathematics of this are summarized in Table 1, from which the updates to P and Q can be read under all possible circumstances. The number γ is a learning rate parameter which regulates the magnitude of the updates the learner makes to the weight distribution; this parameter must satisfy the technical requirement $0 < \gamma < 1$.[3]

Table 1: Learning algorithm for L1 learner (Bush & Mosteller 1955, Yang 2002). The new values of P and Q are $P' = P + \Delta P$ and $Q' = Q + \Delta Q$. A positive ΔP (resp. ΔQ) rewards G_1 (resp. G_2), while a negative value punishes it.

Input string	Grammar chosen	Update to P	Update to Q
$s \in L_1 \setminus L_2$	G_1	$\Delta P = \gamma Q$	$\Delta Q = -\gamma Q$
$s \in L_1 \setminus L_2$	G_2	$\Delta P = \gamma Q$	$\Delta Q = -\gamma Q$
$s \in L_1 \cap L_2$	G_1	$\Delta P = \gamma Q$	$\Delta Q = -\gamma Q$
$s \in L_1 \cap L_2$	G_2	$\Delta P = -\gamma P$	$\Delta Q = \gamma P$
$s \in L_2 \setminus L_1$	G_1	$\Delta P = -\gamma P$	$\Delta Q = \gamma P$
$s \in L_2 \setminus L_1$	G_2	$\Delta P = -\gamma P$	$\Delta Q = \gamma P$

To model feature responsivity, Kauhanen (2022) introduced a simple modification to this basic algorithm. For a learner past the critical age, a second learning rate parameter δ is introduced whose effect is to render the acquisition of G_1 more difficult for the learner.[4] The update rule is given in Table 2. It is obvious that, under this modification of the algorithm, grammar G_1 faces an inherent disadvantage compared to grammar G_2: even when the former is rewarded after parsing success, an amount of weight is deducted (specifically, that amount is

[3]Technically, $\gamma = 1$ is also possible, leading to a model in which learners "hop" categorically between G_1 and G_2 (cf. Gibson & Wexler 1994, Niyogi & Berwick 1996). I leave this special case aside here.

[4]To ensure P and Q always remain probabilities, a technical requirement is that $0 < \delta < 1 - \gamma$.

Table 2: Learning algorithm for L2 learner (Kauhanen 2022). The new values of P and Q are $P' = P + \Delta P$ and $Q' = Q + \Delta Q$.

Input string	Grammar chosen	Update to P	Update to Q
$s \in L_1 \setminus L_2$	G_1	$\Delta P = \gamma Q - \delta P$	$\Delta Q = -\gamma Q + \delta P$
$s \in L_1 \setminus L_2$	G_2	$\Delta P = \gamma Q - \delta P$	$\Delta Q = -\gamma Q + \delta P$
$s \in L_1 \cap L_2$	G_1	$\Delta P = \gamma Q - \delta P$	$\Delta Q = -\gamma Q + \delta P$
$s \in L_1 \cap L_2$	G_2	$\Delta P = -\gamma P - \delta P$	$\Delta Q = \gamma P + \delta P$
$s \in L_2 \setminus L_1$	G_1	$\Delta P = -\gamma P - \delta P$	$\Delta Q = \gamma P + \delta P$
$s \in L_2 \setminus L_1$	G_2	$\Delta P = -\gamma P - \delta P$	$\Delta Q = \gamma P + \delta P$

δP), whereas no such deduction occurs for G_2 (by contrast, to retain the identity $P + Q = 1$, the amount δP is always *added* to the weight of G_2).

For reasons which will become obvious shortly, it is useful to define the quantity

$$d = \delta/\gamma, \tag{5.1}$$

i.e. the ratio of the two learning rate parameters. Note that for a pre-critical-age learner $d = 0$, whereas for a post-critical-age learner $d > 0$ (assuming that G_1 carries some amount of L2-difficulty). The ratio d expresses the degree of L2-difficulty of G_1, normalized by the general learning rate γ. It is thus a general measure of L2-difficulty whose value does not depend on the overall speed with which the learner adjusts the grammar weights P and Q.

Let c_1 denote the probability with which the learner receives an input sequence which G_1 does not accept, and let c_2 similarly denote the probability with which an input sequence not accepted by G_2 is received. These "penalty probabilities" (Yang 2000) determine the ultimate course of learning in the following sense. If it is possible to assume that the values of c_1 and c_2 do not change during the learning period (i.e. the penalty probabilities are constant), we say the learner operates in a *stationary* learning environment. If, moreover, both c_1 and c_2 are non-zero, the learner's weights for G_1 and G_2 converge onto a unique stationary distribution with increasing learning iteration regardless of the weights' initial values (Bush & Mosteller 1955, Narendra & Thathachar 1989). In practice this means that, under this learning algorithm, any learning trajectory can be broken down into two phases, a transient phase during which the values of P and Q are moving towards stationarity, and a stationary phase in which statistical equilibrium has been reached. This is illustrated in Figure 2.

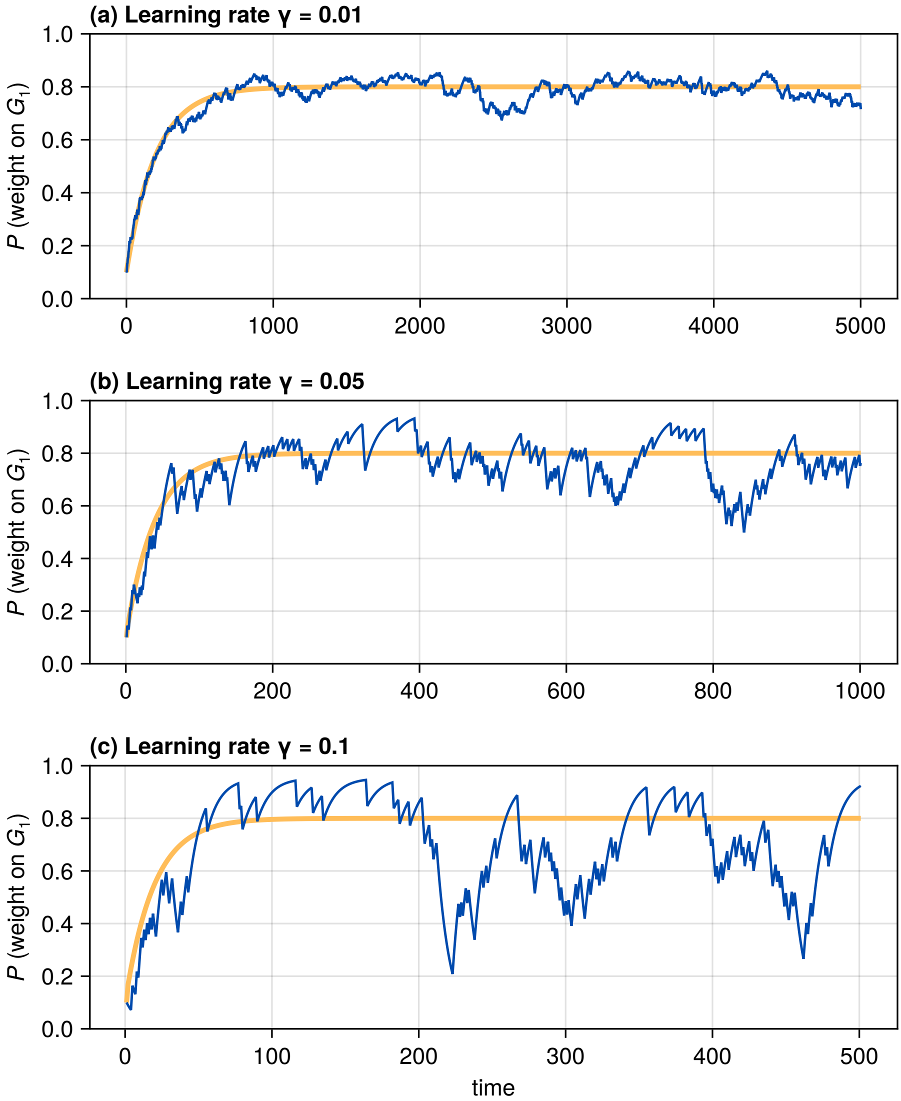

Figure 2: Three simulated learners, each in the same learning environment characterized by penalty probabilities $c_1 = 0.05$ and $c_2 = 0.4$ but with different learning rates: $\gamma = 0.01$, $\gamma = 0.05$ and $\gamma = 0.1$. Each learner additionally is subject to an L2-difficulty of $d = 0.05$. With these choices, the expected value of P at stationarity is 0.8 (the thick yellow curve gives the evolution of the expected value). With the parameters chosen, and with each learner starting from $P(0) = 0.1$, the stationary distribution is reached after about the first $10\gamma^{-1}$ learning iterations.

Since $P + Q = 1$, we may concentrate on P, the weight on the L2-difficult grammar G_1, in what follows. The mean and variance of P at stationarity may be expressed in closed form. Let $\mathrm{E}[P]$ stand for the mean (i.e. expected value) and $\mathrm{Var}[P]$ for the variance of P. At equilibrium,

$$\mathrm{E}[P] = \frac{c_2}{c_1 + c_2 + d} \tag{5.2}$$

and

$$\mathrm{Var}[P] = -\mathrm{E}[P]^2 + \frac{CG^2 + 2c_2G\gamma}{1 - G^2 + 2CG\gamma}\mathrm{E}[P] + \frac{c_2\gamma^2}{1 - G^2 + 2CG\gamma} \tag{5.3}$$

with $C = 1 - c_1 - c_2$ and $G = 1 - \gamma - \delta$ (see Kauhanen 2022: Supplementary file, equations (20) and (24)).

A few remarks are in order. First, once statistical equilibrium has been reached, the expected value of the learner's weight only depends on two things: (i) the penalty probabilities c_1 and c_2, and (ii) the L2-difficulty factor d. In particular, this expected value is not affected by the absolute magnitudes of the learning rate parameters. The variance, on the other hand, is – the higher the learning rates, the higher the variance, and consequently the more fluctuations we observe at equilibrium (Figure 2). This will play a crucial role later when we break away from the deterministic limit.

Secondly, the expected weight on G_1 (the grammar which contains the L2-difficult feature) at equilibrium is always lower for an L2 learner compared to an L1 learner, and the difference is greater the higher the degree of L2-difficulty d is.

Thirdly, although we have not proved it here, the approach to stationarity is exponential, meaning that the absolute difference between the mean of the stationary distribution and the random variable's current value decreases exponentially. This fact (which is also apparent from the simulated trajectories in Figure 2) is important for the following reason: we need not in general worry about whether learners reach equilibrium; convergence is quick. A related fact (again not strictly proven here) is that the learner's initial state does not influence the stationary distribution; the learner is "ergodic" (Narendra & Thathachar 1989).

Most work on the variational learning model has assumed, for the sake of tractability, that the variance does not exist at equilibrium (e.g. Yang 2000, Heycock & Wallenberg 2013, Simonenko et al. 2019, Kauhanen 2022). This allows one to assume that every learner in a population of learners is described by a single number, namely the expectation $\mathrm{E}[P]$. It is then relatively straightforward to write down an inter-generational equation (in either discrete or continuous time) which describes how the weight in one generation of learners is determined from the value of that weight in an immediately preceding generation of learners (under the usual assumptions, this leads to a sigmoidal "S-curve" at the population level; see Wallenberg et al. 2025). Section 3 reviews this strategy particularly as it applies to the modelling of contact-induced change. Section 4 reintroduces the variance; the remaining sections then ask to what extent the predictions made by the deterministic limit continue to hold when populations are not deterministic.

3 The deterministic limit and the bifurcation threshold

In real populations, the penalty probabilities c_1 and c_2 of the competing grammars may differ from learner to learner (depending on which other language users they interact with) and need not remain constant over the duration of learning. Analysing a model that attempts to reproduce this reality incurs significant challenges – the numerous interdependencies between different language users in such a model mean that analytical results are (nearly) unobtainable. It makes sense to start from a simpler vantage point, one which employs a small set of simplifying assumptions that together render the model analytically tractable. This is the approach adopted in Kauhanen (2022), briefly summarized next.

Assume a mixed population of L1 and L2 learners. Every learner internalizes weights P and Q for two grammars, as outlined above. Let x refer to the probability of encountering an L1 learner employing G_1 at a given point in time (mathematically, this is equivalent to the expected value of P, aggregated over the L1 population). Similarly, let y refer to the probability of encountering an L2 learner employing G_1. The state of the population is described by the pair of numbers (x, y); since these are probabilities, the state space is the unit square $[0, 1] \times [0, 1]$. We now ask: what is the eventual fate of the dynamics? Does the state (x, y) tend to some definite value, and if so, what is that value? Moreover, under what conditions do we have $(x, y) \to (0, 0)$, so that no one (the L1 learners included) uses the L2-difficult grammar in the end?

Suppose the following assumptions are made:

1. Learners are arranged in a discrete sequence of non-overlapping generations.

2. Each generation consists of infinitely many learners.

3. Each learner in generation n picks interlocutors from generation $n - 1$ uniformly at random.

4. There is zero variance at any learner's stationary distribution.

Under these assumptions, Kauhanen (2022) shows that (x, y) tends to a global attractor (x^*, y^*) which either is the origin $(x^*, y^*) = (0, 0)$ or lies in the interior of the unit square. In the former case G_1, the grammar with the responsive feature, eventually gives way completely to grammar G_2, which incurs no L2-difficulty. It is important to note that, in this case, the effects of L2-difficulty also percolate to the L1 learner population, even though the latter themselves face no L2-difficulty in learning: the presence of L2 learners shifts the linguistic input L1 learners are exposed to, lowering the probability of them encountering strings which are uniquely accepted by the L2-difficult grammar G_1. Through this mechanism, the effects of L2 learning may percolate through the whole population – at least in principle.

Whether this sort of "total percolation" or "total simplification" (total replacement of G_1 by G_2) occurs depends on whether σ, the proportion of L2 learners in the population, exceeds a critical threshold σ_{crit} whose value is

$$\sigma_{\text{crit}} = \left(1 - \frac{a_2}{a_1}\right)\left(1 + \frac{a_2}{d}\right) \tag{5.4}$$

(Kauhanen 2022: equation 19). Note that the critical threshold is a relation involving four parameters: the proportion of L2 learners (σ), the amount of L2-difficulty experienced by the L2 learners (d), and the advantages of the competing grammars (a_1 and a_2).

Equation (5.4) can be used to make and test predictions about the outcomes of individual language contact situations. One of the empirical case studies considered in Kauhanen (2022) concerns the expression of null subjects in Afro-Peruvian Spanish (APS), a variety of Spanish spoken by descendants of people brought as slaves to Colonial Peru in the 17th–19th centuries. Sessarego (2015) argues that present-day APS exhibits a mixed system of null subjects, analysable as grammar competition between a consistent null subject grammar and a non-null subject grammar.[5] To illustrate, consider the use of overt subjects in (5.5), where Standard Peninsular Spanish would prescribe null subjects.

(5.5) Afro-Peruvian Spanish (Sessarego 2014: 382)

Paco	*fue*	*a*	*casa.*	*Él*	*se*	*tomó*	*una*	*botella*	*de*	*cerveza*
Paco	went	to	home	he	himself	took	a	bottle	of	beer

[5] For more detailed discussion of the null subject phenomenon and a typology of null subject languages, see Chapter ??.

y	*después*	*él*	*se*	*fue*	*al*	*bar*	*de*	*fiesta.*
and	afterwards	he	himself	went	to the	bar	of	party

'Paco went home. He drunk a bottle of beer and then went to the bar to have fun.'

This proliferation of overt subjects is not unique to APS, but rather has been documented in a number of varieties of Spanish which historically were the objects of European colonialism (see Chapter ?? for extended discussion). Importantly, null subjects have been found to incur L2-difficulty in a number of experimental studies (Bini 1993, Pérez-Leroux & Glass 1999, Margaza & Bel 2006), suggesting a plausible psycho-sociolinguistic explanation for the move away from null subjects in these languages. These remarks notwithstanding, APS did not lose null subjects entirely, and (unlike e.g. the genetically related French) is not classified as a non-null subject language. From the point of view of sociolinguistic typology, then, the challenge in this particular empirical case study is to explain why, despite obvious extensive linguistic admixture in Colonial Peru, APS did not fully lose null subjects, but rather landed on a mixed system.

FIXME: can I find an example sentence from Sessarego that illustrates the mixed nature of APS? That would seem to be important in the present context.

Using available demographic data (which is admittedly scant), Kauhanen (2022) estimated the likely proportion of L2 learners at around $\sigma \approx 0.5$, while corpus estimates by Simonenko et al. (2019) put the advantage of the null subject grammar G_1 at $a_1 = 0.7$ and the advantage of the competing non-null subject grammar G_2 at $a_2 = 0.05$. Although we at present lack any empirical estimates of the magnitude of the L2-difficulty parameter d in this (or indeed in any other) case, the critical value σ_{crit} has a lower bound which is attained as $d \to \infty$; this is

$$\lim_{d\to\infty} \sigma_{\text{crit}} = 1 - \frac{a_2}{a_1}, \tag{5.6}$$

as is easy to verify from 5.4. Plugging in the advantage parameters, we find

$$1 - \frac{a_2}{a_1} = 1 - \frac{0.05}{0.7} \approx 0.93. \tag{5.7}$$

Since obviously $0.5 < 0.93$, the prediction that APS settles on a mixed system is thus borne out. In other words, in this case the great formal advantage (proportion of input strings parsed) enjoyed by the null subject grammar in comparison to the non-null subject grammar means that around 93% of the population would have to be learning the language as an L2 for full simplification to take place. The empirical value of the proportion of L2 learners was almost certainly lower than this, and so full simplification is not expected.

Section 7 below will revisit the APS case study. There, in the context of a full stochastic simulation of learning and population dynamics, we will be in a position to examine exactly how far from full simplification the language remained at the moment of greatest influence from L2 learning.

4 Embracing variation

Adopting the simplifying assumptions listed in the previous section made it possible to derive equation (5.4), the bifurcation threshold for contact-induced simplification in the deterministic limit. In relation to real-world application, this result suffers from the following main limitations, which arise immediately from the simplifying assumptions made:

1. It assumes *complete and random mixing*. Everybody can come into contact with everybody else; in particular, there is no sense in which L2 learners might be more likely to interact among each other than with L1 learners (as might have been realistically the case in, for instance, the sorts of colonial settings which gave rise to languages such as APS).

2. It does not allow for *variation between learners in the same (L1/L2) group*. Yet the L2 acquisition literature abounds in empirical data which strongly suggests that some individuals attain higher fluency than others, and even the usage patterns of L1 speakers vary from person to person.

3. It assumes a *constant proportion of L2 learners* σ. However, in several empirically interesting situations (including the history of APS), it is the case that σ first goes up, then comes down again as the children of immigrants acquire the language as an L1 rather than as an L2.

Relaxing these assumptions requires us to view the system as an agent-based model (Gilbert 2020, Smaldino 2023) and to run computer simulations, something that introduces a number of technical challenges. For instance, even though it was pointed out above that the approach to a learner's stationary distribution is exponential, this may still take thousands of learning iterations depending on the specific values of the learning rate parameters and the penalty probabilities. If there are thousands of learners in the population, the numbers quickly multiply, leading to long simulation runtimes. While this is often not a problem from the point of view of running a single simulation, a single simulation outcome is never sufficient to give anything approaching a complete picture of the model's possible behaviours. First, to understand the role played by stochastic factors, the outcomes of multiple simulation runs must be summarized (typically by taking averages). Secondly, and more importantly, the many model parameters involved result in a combinatorial explosion if we wish to sweep over the parameter space in systematic fashion.

To illustrate this, suppose it took one minute to run a single simulation of interest – a reasonable estimate for a reasonably complex model run on a modern computer. If there are three model parameters, each of which can take on 10 different values, then running the simulation for all possible model parameter combinations will take 10^3, i.e. one thousand, minutes. This corresponds to about 16 hours, still a reasonable time to wait perhaps. But now note that, as emphasized above, we may wish to repeat the simulation for each model parameter combination a number of times, say 50 times, to be able to assess the effect of stochastic factors. Unless it is possible to parallelize the simulation code, this immediately raises the runtime to 830 hours, or just over one month. Adding just one more model parameter to the sweep (assuming again 10 different possible values for the parameter) multiplies the runtime by 10, requiring us to wait 10 months for our results. Requiring more resolution of the parameter sweep (e.g. 100 possible values for each parameter) is clearly impossible in practice, unless we have access to a supercomputer.

It is here that the availability of some analytical results becomes very useful. As explained in Section 2, in a stationary learning environment the behaviour of learners may be summarized by the properties of the distribution of P at statistical equilibrium. Although the exact distribution is unknown, its mean and variance can be computed. It turns out that these are enough to characterize that distribution in practice.

Specifically, we adopt the following strategy – which could be termed the "Beta approximation" – in the following simulations.[6] Instead of simulating each learner's learning trajectory up to statistical equilibrium, we sample the learner's ultimate value of P from a Beta distribution with suitable shape parameters α and β. The shape parameters are obtained via a method known as moment matching.

[6]The Beta distribution is a continuous probability distribution over the interval $[0, 1]$ and is thus a suitable choice to describe the distribution of a random variable such as our P. It is parameterized by two parameters, $\alpha > 0$ and $\beta > 0$, which determine the distribution's shape (see e.g. Samaniego 2014: 151–152). For a different kind of application of a Beta approximation in linguistics (in the context of a Wright–Fisher model of language change), see Guerrero Montero et al. (2025).

The mean and variance of the Beta distribution are

$$\mu = \frac{\alpha}{\alpha + \beta} \tag{5.8}$$

and

$$\sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} \tag{5.9}$$

respectively (see e.g. Samaniego 2014: 151). Thus, to obtain a Beta distribution with a desired mean and variance, we simply need to solve the above equations for the shape parameters α and β, and plug in the mean and variance. But since we know the mean and variance of the learner's stationary distribution, we may now use these numbers to obtain a Beta distribution that approximates that stationary distribution.

To illustrate, Figure 3 shows the results of a number of experiments; model parameters were drawn at random and are listed in Table 3. In each experiment, 10,000 learners were simulated up to statistical equilibrium. The histograms depict that equilibrium, i.e. the distribution of P across all 10,000 learners.[7] The curves give the probability density of the Beta distribution found through the method of moment matching. It is evident that, in each case (i.e. for different values of the model parameters c_1, c_2, γ and δ), the agreement is very good, justifying the use of the Beta approximation.

Table 3: Model parameters (c_1, c_2, γ and δ; randomly drawn) for the simulation experiments of Figure 3, together with the mean (E[P]) and variance (Var[P]) of the stationary distribution (5.8–5.9) and the moment-matched shape parameters (α, β) of the approximating Beta distribution.

	c_1	c_2	γ	δ	E[P]	Var[P]	α	β
(a)	0.94	0.66	0.027	0.083	0.14	0.0008	22.6	137.6
(b)	0.87	0.73	0.012	0.073	0.1	0.0002	45.3	431.1
(c)	0.34	0.31	0.081	0.038	0.28	0.0091	5.8	15.2
(d)	0.14	0.57	0.085	0.09	0.32	0.0058	11.8	24.9
(e)	0.32	0.09	0.1	0.033	0.12	0.0093	1.3	9.2
(f)	0.25	0.87	0.082	0.013	0.68	0.0056	25.5	12.0

[7] In fact, since the learning scheme is ergodic and each learner within a given experiment faces the same learning environment (the same penalty probabilities c_1 and c_2) and employs the same learning rate parameters, the same result could be obtained by sampling 10,000 values from the stationary distribution of a single learner.

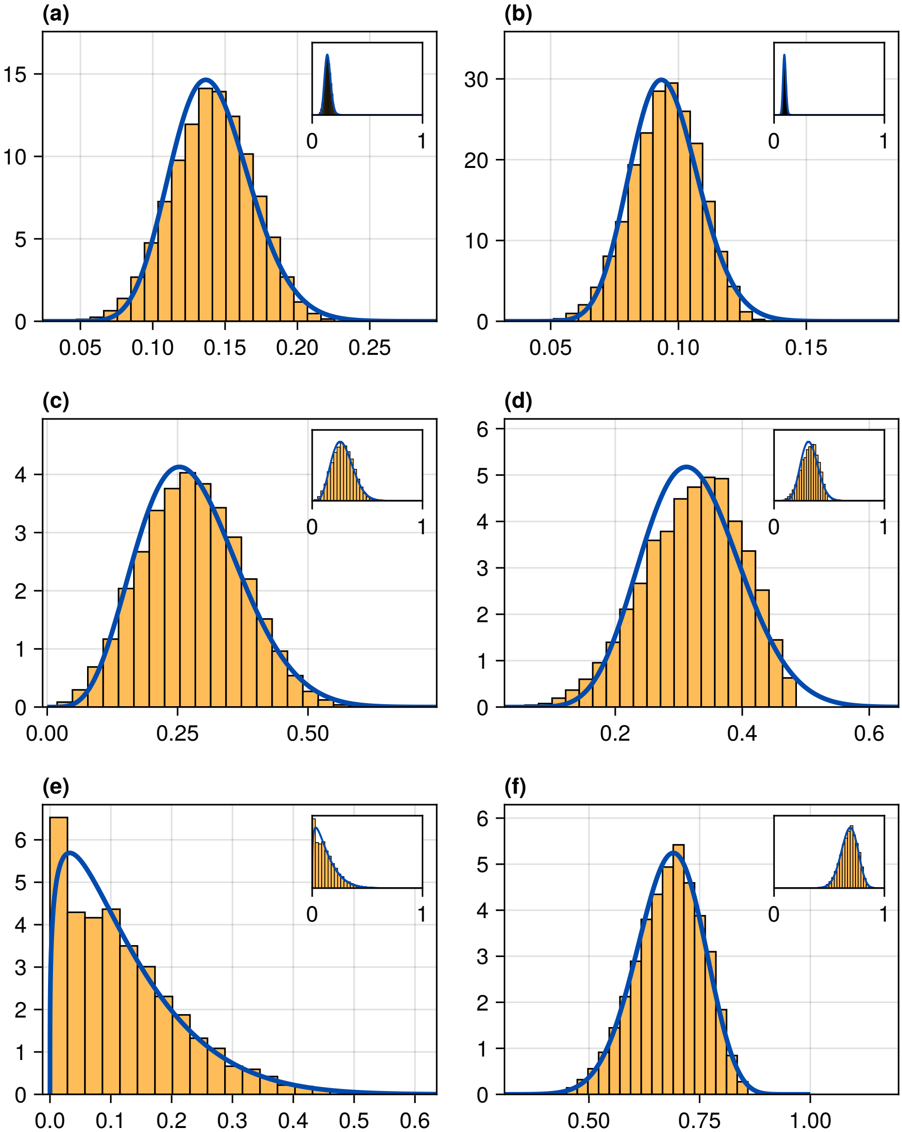

Figure 3: Six learning simulation experiments. In each experiment, 10,000 learners were simulated to equilibrium with randomly drawn parameters (see Table 3). The equilibrium distribution of P, the weight on G_1 (yellow histogram), is well approximated by a Beta distribution (blue curves indicating the probability density function) found by moment matching. Insets scale each facet to the entire range of possible values of P, the interval $[0, 1]$.

As a consequence, simulation runtimes are drastically reduced. On the computer used to simulate the above experiments,[8] the median runtime for simulating one learner to equilibrium was on the order of five hundred microseconds. By contrast, sampling from the Beta distribution only took around four hundred nanoseconds on average, meaning that the latter operation is over a thousand times (more than three orders of magnitude) faster.

5 Simulation setup

For the agent-based simulations, the following setup is assumed. Simulations are carried out on a spatial substrate which is a regular two-dimensional lattice of $L_h \times L_v$ sites with non-periodic[9] boundaries, L_h giving the horizontal dimension and L_v the vertical dimension. While L1 learning may occur in any part of this space, L2 learning is constrained to occur only in the lower half of the lattice (i.e. in sites with a vertical index less than half of L_v). The shape of the lattice is varied through the use of an *aspect ratio* parameter ρ, defined as the ratio of lattice width to lattice height. For $\rho > 1$, the lattice is wider than it is tall, and for $\rho < 1$, it is taller than it is wide (see Figure 4).[10]

The decision to constrain L2 learning to the lower half of the lattice, and the use of the aspect ratio parameter ρ, together allow us to control the intensity of the initial interactions between L1 and L2 learners, i.e. interactions between the top and bottom halves of the lattice. This structure may be interpreted either as spatial (as in many situations of language contact between two contiguous geographical regions) or as social (as in situations of colonization and slavery) in origin. In a very rudimentary way, this will allow us to probe different kinds of scenarios of language contact (cf. Muysken 2013).

Simulations proceed in discrete time, each time step interpreted as one year of real time. At each time step, three actions are performed:

1. Birth and L1 acquisition. A number of speakers are created and made to undergo the process of L1 acquisition. The number of speakers "born" this way is proportional to (i) a birth rate parameter, β, (ii) the current population size, N,

[8] A PC with an Intel Core i5-13600KF processor and 64 GB of DDR4 RAM. All simulation code was written in Julia (Bezanson et al. 2017), version 1.11.4, and can be obtained from https://github.com/erc-starfish/contact-simulations.

[9] This means that the lattice does not "wrap around" at its edges.

[10] Concretely, the horizontal and vertical dimensions are set as $L_h = \lfloor\sqrt{\rho L}\rfloor$ and $L_v = \lfloor\sqrt{\rho^{-1} L}\rfloor$ where L is the approximate desired total number of sites on the lattice.

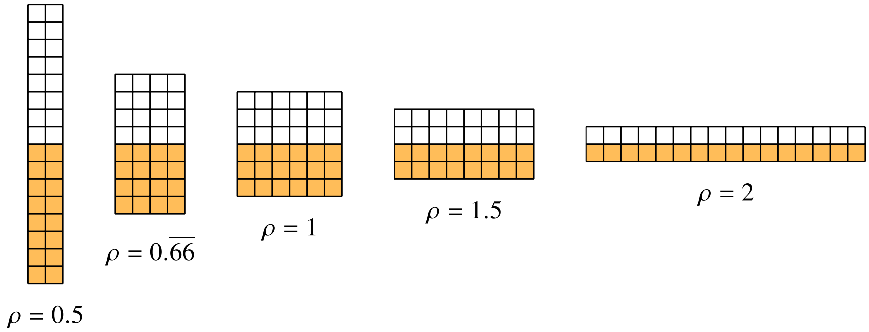

Figure 4: The aspect ratio parameter ρ is used to set the relative horizontal and vertical dimensions of the lattice. L2 learning is constrained to occur only in the bottom half of the lattice (yellow), and so ρ implicitly controls the length of the interface between the initial L1 and L2 learning regions. (Lattices in the actual simulations are far larger, with a total of $L \approx 5000$ sites.)

and (iii) a carrying capacity, K; the latter ensures that population sizes do not explode exponentially (which would lead to computationally intractable, and also empirically unrealistic, simulation runs). Concretely, the number of newborn speakers is sampled from a binomial distribution with success probability β and number of trials $\lfloor N(1-(N/K))\rfloor^+$, where $\lfloor x\rfloor^+ = \lfloor x\rfloor$ (i.e. the integer part of x) if $x \geq 0$ and $\lfloor x\rfloor^+ = 0$ if $x < 0$. Each newborn speaker is assigned to a lattice site at most R sites away from the site occupied by a randomly chosen already existing speaker, the latter interpreted as the newborn's parent. Multiple speakers may occupy the same site; once placed on the lattice, a speaker stays in the same site for its lifetime. L1 acquisition is modelled by sampling from a Beta distribution as explained in Section 4. The learner estimates the penalty probabilities c_1 and c_2 from its local environment. Concretely, the learner samples F speakers at random from within a distance (Chebyshev distance) of R on the lattice, or takes all neighbouring speakers if their number does not exceed F. The local penalty probabilities c_1 and c_2 are then calculated from the frequencies of use of grammars G_1 and G_2 among these neighbours, together with the advantage parameters a_1 and a_2, in the usual manner: if x stands for the local frequency of G_1, then $c_1 = a_2(1-x)$ and $c_2 = a_1 x$ (Yang 2000, Wallenberg et al. 2025).

2. Immigration and L2 acquisition. A number of speakers are "imported" into the lattice and made to undergo the process of L2 acquisition. The number of immigrants is, like the number of newborn agents in step 1, dependent on both the current population size and the carrying capacity, and is similarly sampled from a binomial distribution, except that a migration rate parameter τ is used instead of the birth rate β. This way of arranging the immigration process is modelled on the assumptions that (i) larger populations attract more immigration, and (ii) a finite substrate can support only a finite amount of immigration. L2 acquisition proceeds as L1 acquisition, via the Beta approximation, except that the learner is also subject to the L2-difficulty parameter d, set as a global parameter for all learners.

In Section A, it is shown that given the above general assumptions about population dynamics, the expected proportion of L2 learners in the population is

$$\mathrm{E}[\sigma] = \frac{\tau}{\beta + \tau} \tag{5.10}$$

once the total population size has stabilized. In other words, this quantity may readily be computed from the birth and immigration rates β and τ. Conversely, in order to obtain a desired expected proportion of L2 speakers $\mathrm{E}[\sigma]$ having set some fixed birth rate β, we may invert the above equation to verify that the immigration rate τ needs to be chosen so that

$$\tau = \frac{\mathrm{E}[\sigma]}{1 - \mathrm{E}[\sigma]}\beta. \tag{5.11}$$

In the following simulations, this is what we do: i.e. rather than directly set the immigration rate τ, we set the desired proportion of L2 learners $\mathrm{E}[\sigma]$ and calculate τ using (5.11).

3. Death. When agents are introduced to the population, they are assigned an age which is 0 for L1 learners and a random number for L2 learners; this random number is constructed by drawing a real number between 0 and an upper limit λ from a truncated normal distribution with mean m and standard deviation s, and rounding down to an integer. Agents are also assigned a lifetime, sampled from a Weibull distribution[11] with shape parameter k and scale parameter λ. At each simulation iteration, the age of every agent on the lattice is incremented by one, and agents whose ages exceed their lifetimes are removed from the lattice. The

[11]The Weibull distribution is a standard choice for modelling the probability of time-until-failure, whether in engineered or biological systems (see Samaniego 2014: 155–156).

particular choices of k and λ used in the simulations (see Table 4) imply that the survival distribution is left-skewed, meaning that death is more likely the older a speaker becomes.

Figure 5 gives example illustrations of the distributions governing the above-described dynamics.

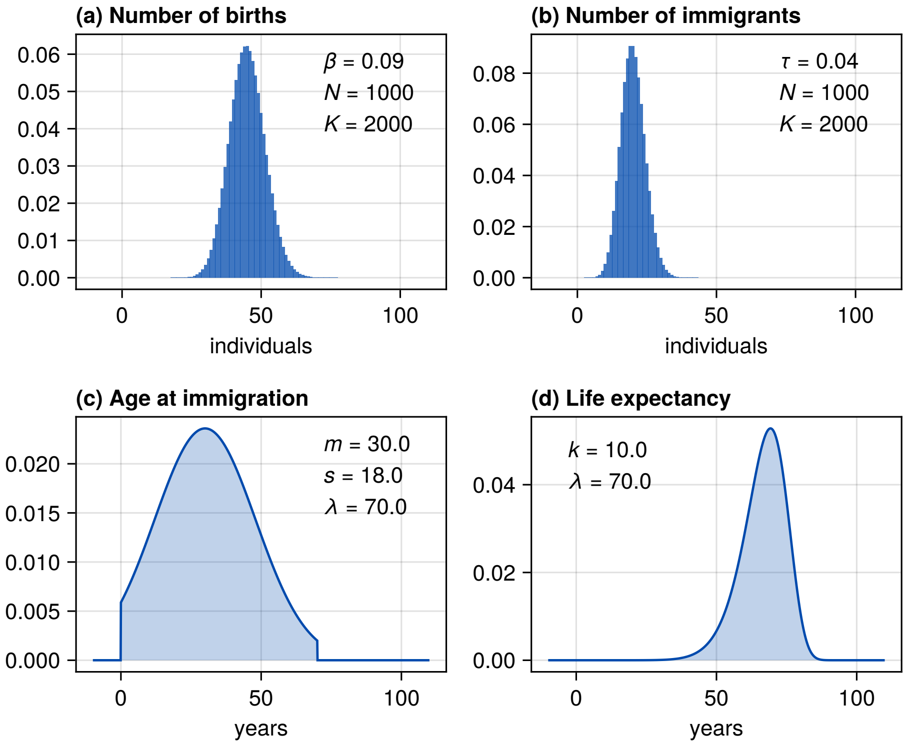

Figure 5: Probability mass (a–b) or density (c–d) functions of example distributions for (a) number of births per year, (b) number of immigrants per year, (c) age of immigrants, and (d) life expectancy of individuals, given the parameters in the plot insets.

Each simulation is initialized with 100 L1 speakers with full use of G_1 ($P = 1$) inserted in random locations of the L1 portion of the lattice. From here, the simulation proceeds according to the above description, with the exception that for the first 50 years, there is no immigration. This is to ascertain that full use of G_1 is in fact a stable population equilibrium in the absence of L2 learning. After the first 50 years, immigration is enabled and continues for the following T_{L2} years.

This parameter allows us to control the temporal extent of the immigration process, facilitating the modelling of both situations of temporally extended moderate bilingualism and more radical population dynamics such as those of colonial societies, which are characterized by strong initial immigration followed by its levelling off at abolition. Each simulation is run for a total of T time steps. The main observable of interest is the population average of P, i.e. the mean usage of the L2-difficult grammar G_1, throughout and, in particular, at the end of the simulation.

It is important to emphasize that while L2 acquisition is constrained to only occur in the lower half of the lattice, no such constraint is in place for L1 acquisition. Thus, when L2 learners have children (i.e. when newborn speakers are placed in the vicinity of existing speakers in the lower half of the lattice), those children undergo their acquisition process as L1 learners.

All model parameters are summarized in Table 4. Parameters which were held constant across all simulations reported in this and the following section are indicated with those constant values. Parameters with values labelled 'various' were varied across simulations; further information on these will be given below.

Figure 6 illustrates a typical simulation run, plotting the mean weight on G_1, the proportion of L2 learners, and normalized population size (the number of speakers divided by its maximum value over the duration of the simulation) against time, for the simulation parameters listed in Table 5. As L2 learners are introduced into the population from time 50 onwards, we observe a steady decrease in the mean weight on G_1. Concomitantly, the proportion of L2 learners increases with the influx of these learners. Ultimately, the weight on G_1 tends toward zero; this aligns with the prediction of the deterministic model, as the proportion of L2 learners exceeds the critical threshold σ_{crit} given the parameter values of Table 5. The L2-difficult grammar is not recovered, at least not instantly, even when the supply of L2 learners is cut off later.

Figure 7 provides an alternative view of the same simulation run. Each of the six panels of this figure provides a snapshot of the lattice at a different point in time, with crosses indicating speakers (upright crosses = L1 learners, slanted crosses = L2 learners; red = higher weight on G_1, blue = lower weight on G_1). To promote legibility, a random sample of 1% of all speakers is drawn, and at slightly jittered locations.

6 Systematic parameter sweeps

In all, the simulation setup described in the previous section involves 18 model parameters. If one were to carry out a systematic parameter sweep across this

Table 4: Overview of model parameters for the agent-based simulations in Sections 5–6.

Parameter	Symbol	Value(s)
Advantage of grammar G_1	a_1	various
Advantage of grammar G_2	a_2	various
Learning rate	γ	various
L2-difficulty of G_1	d	various
Lattice aspect ratio	ρ	various
Approximate lattice size (number of sites)	L	5000
Interaction radius	R	10
Maximum interaction density (number of contacts)	F	50
Birth rate	β	0.05
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$	various
Length of immigration period	T_{L2}	various
Mean L2 learner age at immigration	m	20
S.D. of L2 learner age at immigration	s	10
Number of L1 speakers at time 0	N_0	100
Shape parameter of death distribution	k	10.0
Scale parameter of death distribution	λ	70.0
Population carrying capacity	K	various
Length of simulation (steps)	T	various

Table 5: Model parameters for the simulation depicted in Figures 6–7. For the values of the constant parameters, consult Table 4.

Parameter	Symbol	Value(s)
Advantage of grammar G_1	a_1	0.12
Advantage of grammar G_2	a_2	0.1
Learning rate	γ	0.01
L2-difficulty of G_1	d	10
Lattice aspect ratio	ρ	1.5
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$	0.7
Length of immigration period	T_{L2}	200
Population carrying capacity	K	20000
Length of simulation (steps)	T	500

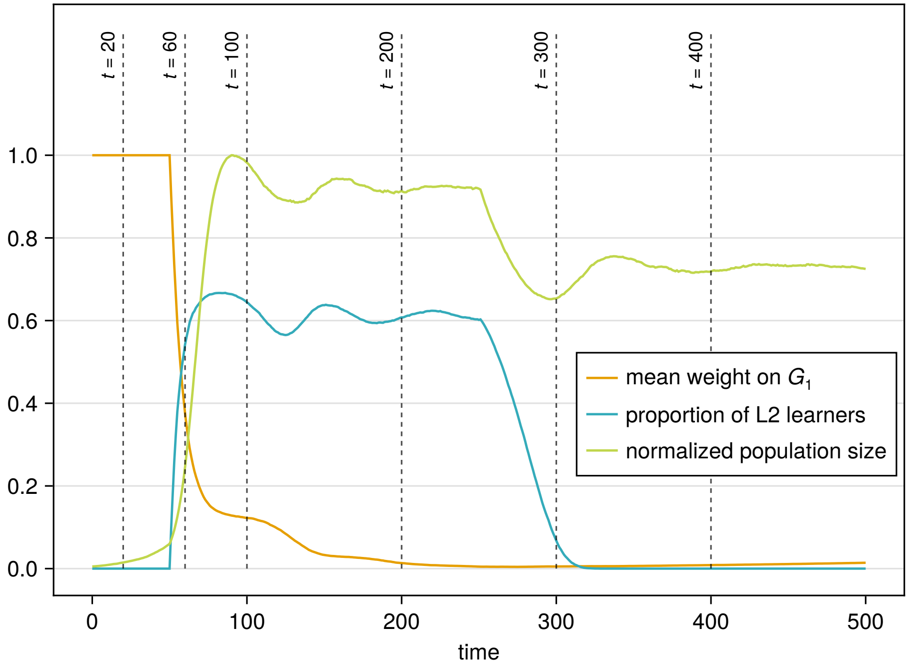

Figure 6: One simulation run. For model parameters, see Table 5. Snapshots of the population at the indicated time points are provided in Figure 7.

parameter space with, say, 10 values per parameter and 50 repetitions for each parameter combination, the number of simulation runs would equal 50×10^{18}. This is intractable even with the Beta approximation. Some insight is thus required in identifying what parameters are likely to be of interest. This, in turn, requires us to formulate specific research questions.

In what follows, I will focus on the following two questions:

1. Does the bifurcation threshold identified in the deterministic limit (5.4) retain its validity in this stochastic simulation setup?

2. How do language contact outcomes depend on (i) interaction patterns between the subpopulations (i.e. on lattice aspect ratio ρ) and (ii) the length of the period of language contact (i.e. the length of immigration parameter T_{L2})?

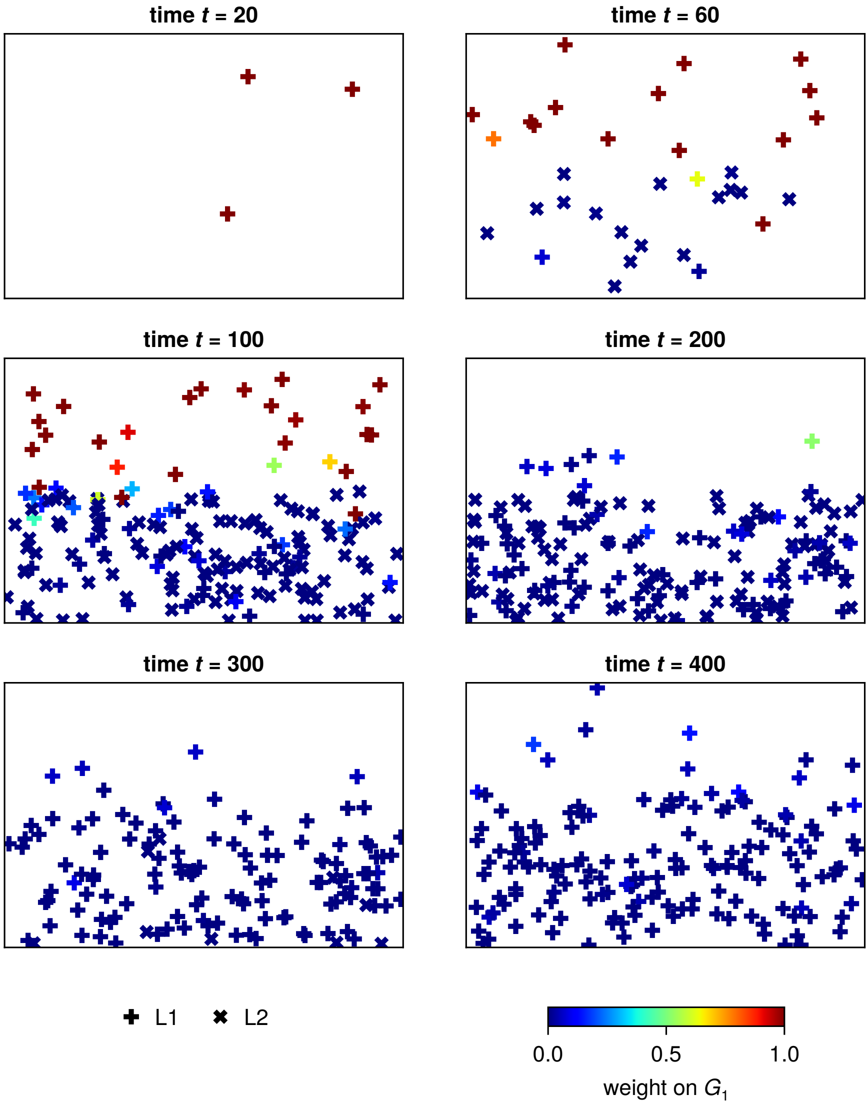

Figure 7: Snapshots from a simulation with the parameters of Table 5. Each cross represents one speaker in its position on the lattice (a random sample of 1% of all speakers is actually visualized, and speaker positions are slightly jittered, for reasons of legibility). Upright crosses represent L1 learners, slanted crosses L2 learners. Colour corresponds to the speaker's weight on the L2-difficult grammar G_1.

To probe question 1, a sweep was carried out by varying the parameters $\mathrm{E}[\sigma]$, the expected proportion of L2 learners, and d, the extent of L2-difficulty on grammar G_1 (see Table 6 for model parameters). For each combination of $\mathrm{E}[\sigma]$ and d, 50 simulations were carried out, each simulation lasting 3000 time steps (i.e. 3000 years of real time). As explained above, L2 learning was enabled from time step 50; moreover, in this particular sweep, the supply of L2 learners was never cut off.

For each combination of proportion of L2 learners σ and L2-difficulty d (and advantage parameters a_1 and a_2; see Table 6), equation (5.4) predicts a specific threshold σ_{crit}. This threshold is depicted in Figure 8 as the thick dashed line. The heatmap provides the eventual value of P (weight on G_1), averaged over the population and over all 50 simulation runs. Furthermore, contour lines are drawn for P across this heatmap. We find that the stochastic simulations are in general agreement with the theoretical (deterministic-limit) prediction, with the following difference: the theoretical calculation appears to somewhat overestimate the extent of simplification. In other words, the contour line closest to the theoretically predicted σ_{crit} is 0.01, corresponding to 1% use of G_1 across the entire population. This is likely due to the presence of local influences in the stochastic model: in contrast to the deterministic-limit prediction, which assumes that every speaker comes in contact with every other speaker equally frequently, in the stochastic simulation here variants may survive in local "pockets" for considerable lengths of time.

Turning to question 2, a second sweep was performed for varying lattice aspect ratios ρ and varying durations of the L2 learning period T_{L2} (for all model parameters, see Table 7). Again, 50 simulations were carried out for every simulation combination, for 3000 time steps. Figure 9 visualizes the results, with time step on the horizontal axis, aspect ratio ρ on the vertical axis, and T_{L2} varying across the panels. We observe an interaction between ρ and T_{L2} – for short periods T_{L2}, tall lattices ($\rho < 1$) conserve the simplification longer than do wide lattices ($\rho > 1$). With higher T_{L2}, the effect effectively disappears.

This interaction with short periods of L2 learning may seem counterintuitive at first: wide lattices ($\rho > 1$) are precisely those that have a longer active interface between the initial L1 and L2 subpopulations. One might thus expect such lattices to favour simplification. However, the active interface works in both directions, i.e. a longer interface also promotes the (anti-simplificatory) effect that L1 speakers have on L2 learners. A taller lattice, by contrast, gives rise to an extended domain in the lower half of the lattice in which simplification can proceed to completion, with the children of the L2 learners acquiring the simplified grammar as an L1.

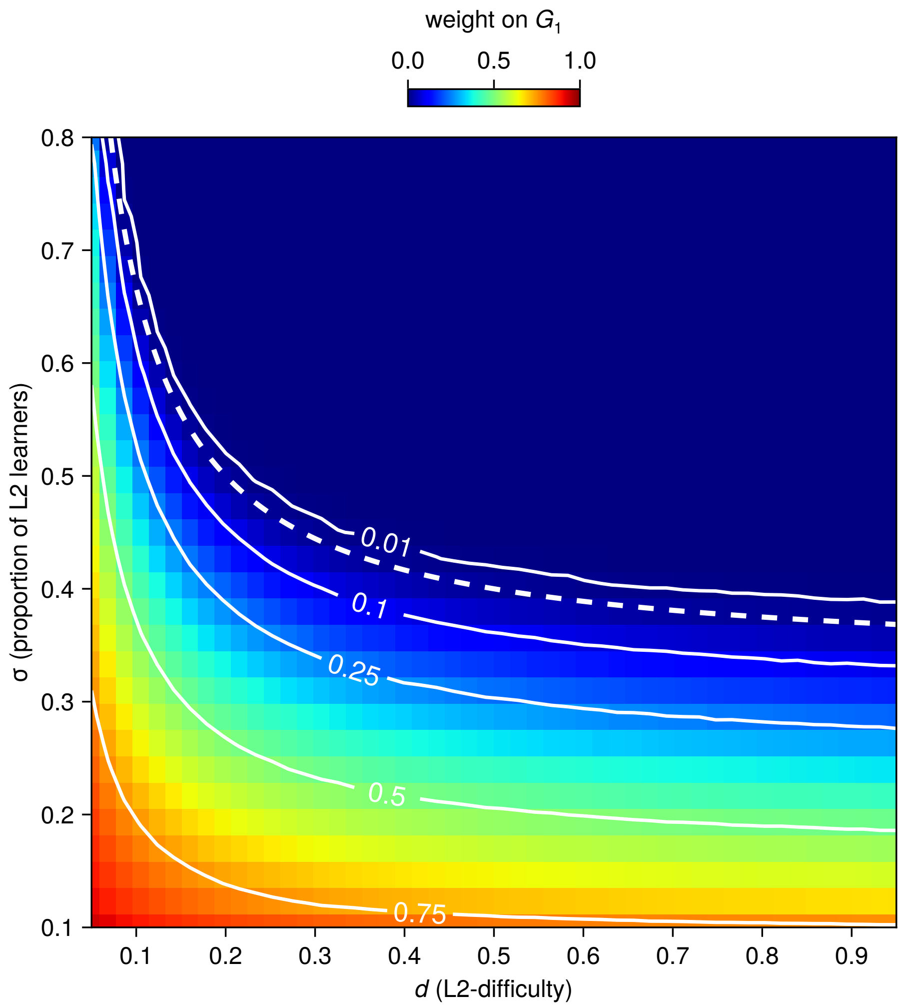

Figure 8: Weight on the L2-difficult grammar G_1 in the long run, as a function of various choices of L2-difficulty d and proportion of L2 learners σ (actual, not expected, proportion recorded at the end of simulation). The thick dashed curve gives the critical threshold (5.4) predicted by the deterministic model. Solid curves are contour lines. For simulation parameters, see Table 6.

Table 6: Model parameters for the sweep of Figure 8. For the values of the constant parameters, consult Table 4. In this and subsequent tables, the notation $[a \overset{n}{\dots} b]$ refers to a sequence of n equally-spaced values from a to b (endpoints inclusive). The notation $\log_{10}[a \overset{n}{\dots} b]$ refers to such a sequence with logarithmic spacing.

Parameter	Symbol	Value(s)
Advantage of grammar G_1	a_1	0.15
Advantage of grammar G_2	a_2	0.1
Learning rate	γ	0.01
L2-difficulty of G_1	d	$[0.05 \overset{50}{\dots} 0.95]$
Lattice aspect ratio	ρ	1.0
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$	$[0.1 \overset{500}{\dots} 0.9]$
Length of immigration period	T_{L2}	2950
Population carrying capacity	K	1000
Length of simulation (steps)	T	3000

Table 7: Model parameters for the sweep of Figure 9. For the values of the constant parameters, consult Table 4.

Parameter	Symbol	Value(s)
Advantage of grammar G_1	a_1	0.12
Advantage of grammar G_2	a_2	0.1
Learning rate	γ	0.01
L2-difficulty of G_1	d	10
Lattice aspect ratio	ρ	$\log_{10}[0.001 \overset{21}{\dots} 1000.0]$
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$	0.7
Length of immigration period	T_{L2}	$[100 \overset{6}{\dots} 600]$
Population carrying capacity	K	10000
Length of simulation (steps)	T	3000

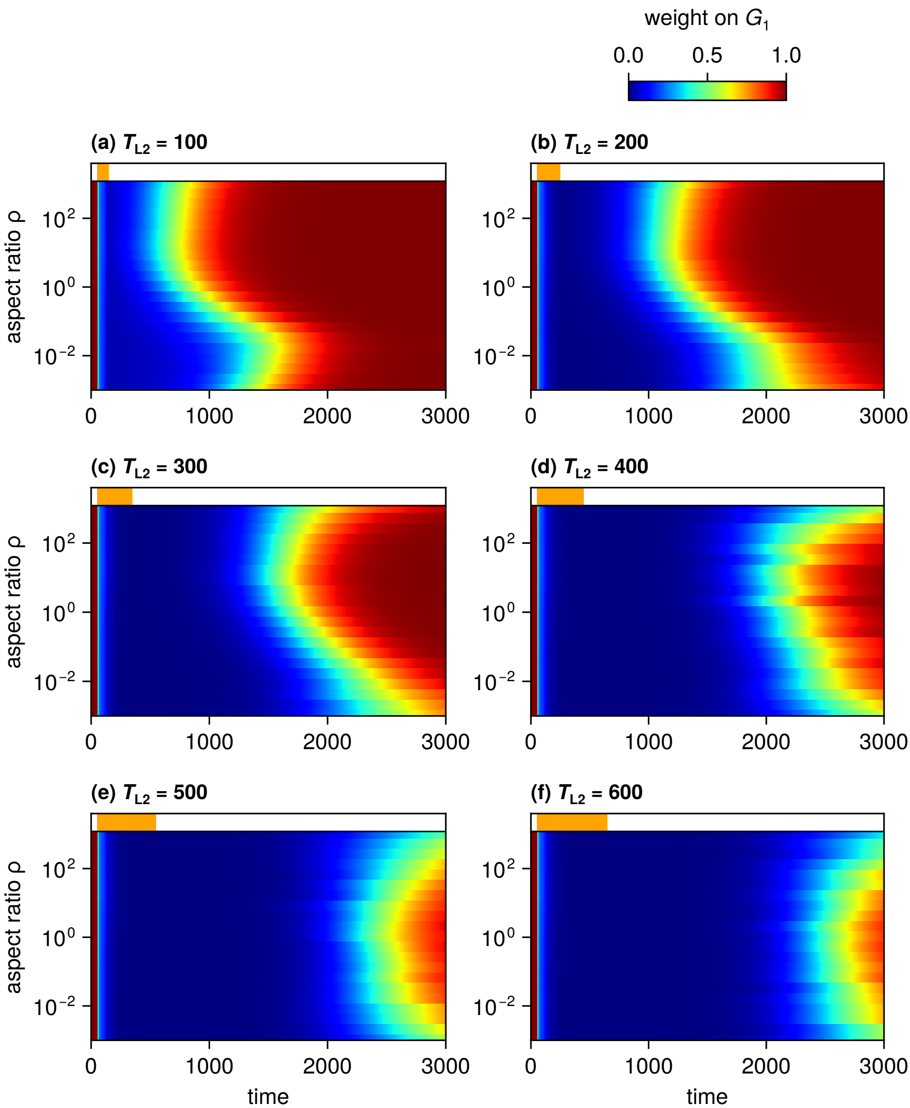

Figure 9: Results of a parameter sweep crossing lattice aspect ratio (ρ) and the length of the period of L2 learning/immigration (T_{L2}). The horizontal axis in each facet represents simulation time step; the period of L2 learning is depicted by the orange bar at the top of each panel.

7 Afro-Peruvian Spanish and Afrikaans revisited

To further test the model, we can ask to what extent its specific predictions on the two case studies considered in Kauhanen (2022) hold, in the light of what is known empirically in both cases. The two case studies are null subjects in Afro-Peruvian Spanish (already discussed above) and verbal inflection in Afrikaans. These two cases not only correspond to different sociohistorical circumstances, they also involve different linguistic properties and different outcomes. As discussed above, the move from null subjects in the direction of overt subjects in APS is only partial; present-day APS has not simplified the L2-difficult null subject grammar completely, and younger generations of speakers of APS in fact use null subjects more than older generations (Sessarego 2014). Verbal inflection in Afrikaans, however, was completely simplified, in the sense that, today, all regular verbs only possess one syncretic form across the paradigm of person and number (see 5.12, and Ponelis 1993 for a detailed treatment).

(5.12) (a) Dutch *lopen* 'to run/walk'

	singular	plural
1st	*loop*	*lopen*
2nd	*loopt*	*lopen*
3rd	*loopt*	*lopen*

(b) Afrikaans *loop* 'to walk'

	singular	plural
1st	*loop*	*loop*
2nd	*loop*	*loop*
3rd	*loop*	*loop*

A successful computational model of these dynamics must therefore be able to predict both developments: the "failure" to simplify in one case, and the "success" in the other.

Such empirical evaluation is tricky, however, due to the many parameters involved and the difficulty in their estimation. Several of the sociodynamic parameters are simply unknown, and there is no conceivable way of retrieving the required information from an empirical source (such is the case of, for example, the mean age of L2 learners at immigration, m; cf. Table 4). With the help of a model, however, these parameters can be estimated (with some uncertainty) from the available historiographic data, which pertains to known numbers of Europeans and non-Europeans both in the Dutch Cape Colony and in Colonial Peru at the relevant time periods. Tables 8–9 give these numbers, slightly simplified from the tables in Kauhanen (2022), which in turn were obtained from the demographic studies in Bowser (1974) and Giliomee & Elphick (1979).

Table 8: Demographic data for Afro-Peruvian Spanish (from Bowser 1974 by way of Kauhanen 2022). The category "mixed", included in the original counts in Bowser (1974), has been dropped as it is unclear whether people of mixed background in the censuses ought to be counted as European or not.

Year	Europeans	Non-Europeans
1600	7193	7059
1614	11867	12364
1619	9706	13403
1636	10758	15046

Table 9: Demographic data for Afrikaans (from Giliomee & Elphick 1979 by way of Kauhanen 2022). Numbers for the indigenous (Khoekhoe) population are only available for years 1798 and 1820; as these numbers cannot be reliably imputed for the earlier years, the indigenous are not included in the numbers shown here.

Year	Europeans	Non-Europeans
1670	125	65
1690	788	429
1711	1693	1834
1730	2540	4258
1750	4511	5676
1770	7736	8572
1798	20000	27454
1820	42975	33711

To fit the mathematical model to these data, I use Approximate Bayesian Computation Sequential Monte Carlo (ABC-SMC) to obtain posterior distributions for the values of all sociodemographic parameters of the model (**REF**). Briefly put, this method starts from a prior distribution for each model parameter (here, uniform priors were used). A combination of parameter values is sampled from the prior distributions and the model is run. The model's predicted numbers of L1 and L2 speakers are then compared to the empirical values by computing the difference between predictions and empirical values. A new proposal of model parameter values is then obtained by the method of Differential Evolution (Storn & Price 1997, Das & Suganthan 2011) and the model is run again. This procedure is repeated ~~for~~ thousands of times, and those parameter value combinations which result in the smallest distances between the predicted and empirical speaker numbers are retained and form the posterior parameter value distributions (see Section B for further technical details). The posterior distributions ~~so~~ found are illustrated in Figure 10 and Figure 11.

Next, 10 combinations of model parameters were sampled from the posterior distributions to obtain simulations of each case study. These sociodynamic parameters were combined with linguistic parameters: a_1 and a_2, the advantages of the competing grammars, and d, the amount of L2-difficulty on grammar G_1. For the advantage parameters, estimates exist: as mentioned above, we have $a_1 = 0.7$ and $a_2 = 0.05$ in the case of APS by the method of Simonenko et al. (2019). In the case of Afrikaans, Kauhanen (2022) takes $a_1 = a_2 = 1$ since there is no overlap between the forms produced by the two grammars. For the L2-difficulty parameter d, no estimates currently exist, and so we are forced to sweep over this parameter, observing how variation in d affects the model outcomes.

Figure 12 visualizes the mean value of P, the weight on the L2-difficult grammar G_1, for each year from the beginning of the simulation up to present day. Each panel of the figure contains 150 curves, corresponding to five independent simulations for three different values of the L2-difficulty parameter d for each of the ten posterior samples.

For Afro-Peruvian Spanish, the simulation outcomes are consistent with both the empirical reality and the deterministic result from Kauhanen (2022). Firstly, full simplification is never attained, no matter how large the L2-difficulty d. Secondly, once L2 learning is removed, the representation of the L2-difficult grammar G_1 slowly creeps back toward full usage, replicating the observation that younger speakers of APS are reverting back to using a consistent null subject grammar (Sessarego 2014).

For Afrikaans, the simulation results are also roughly of the right shape. For sufficiently high L2-difficulties, full simplification is predicted by the 20th century, apart from a few isolated simulation runs in which stochastic fluctuations cause the weight on G_1 to rise again after an initial decrease. For very low L2-difficulties, simplification takes so long that the process may in fact be halted, if the supply of L2 learners is cut off before full simplification has been attained.

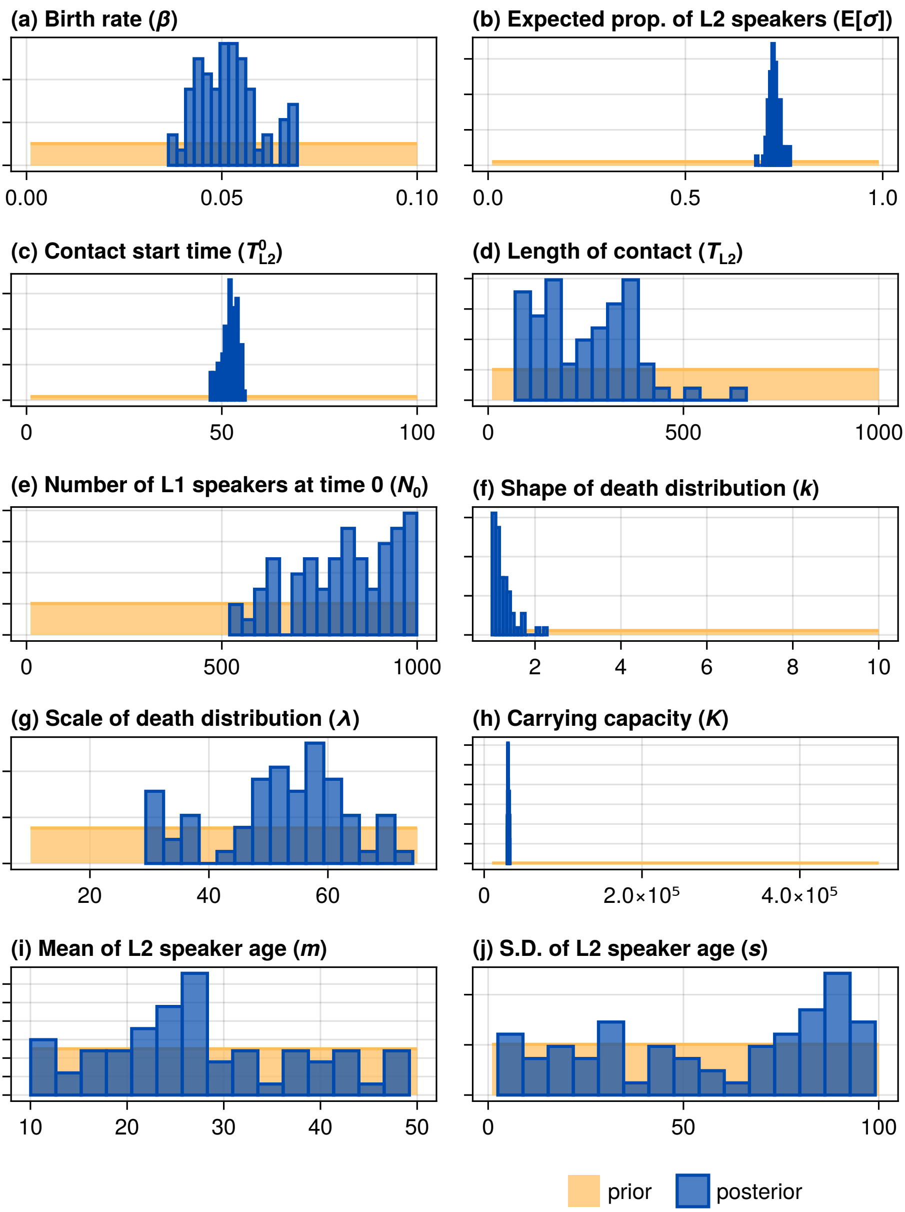

Figure 10: Parameter estimation for Afro-Peruvian Spanish using Approximate Bayesian Computation.

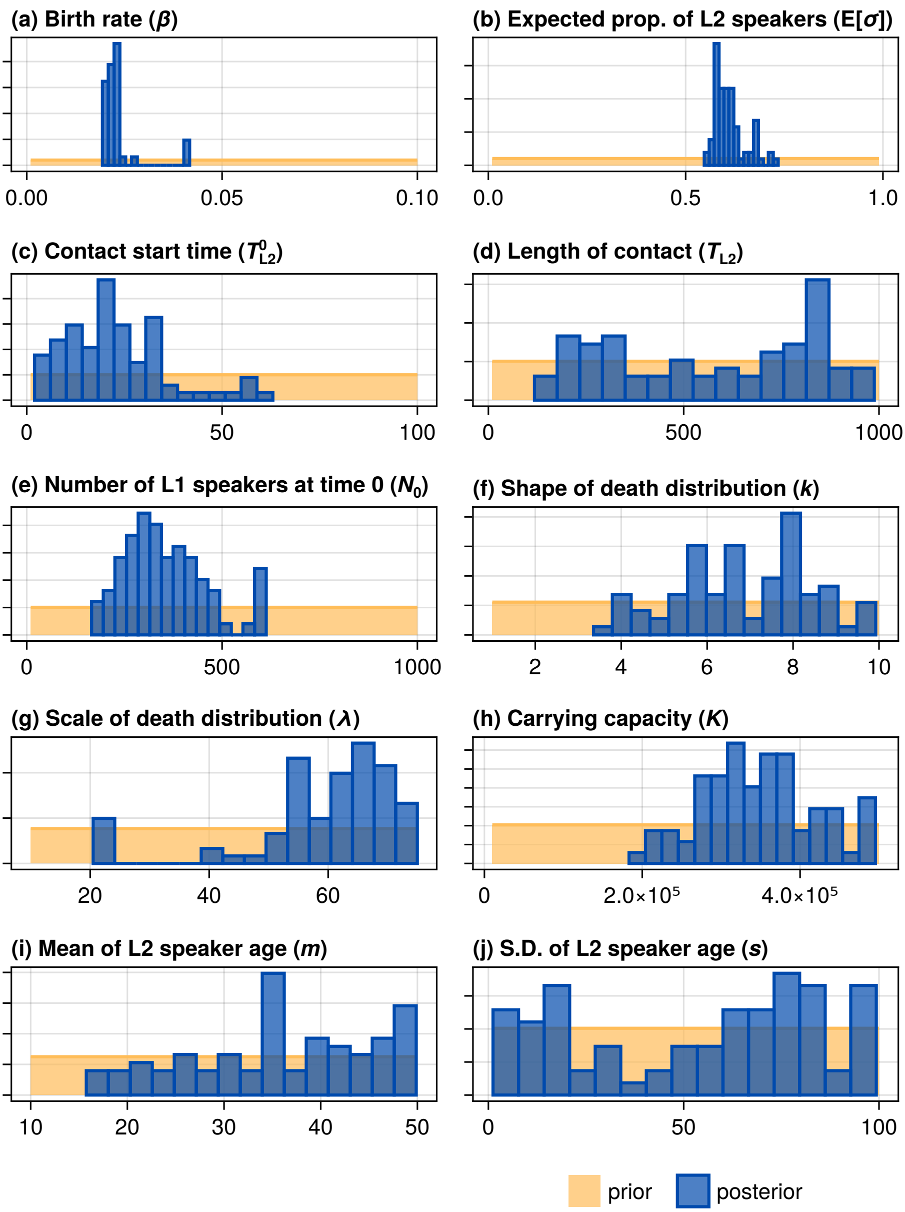

Figure 11: Parameter estimation for Afrikaans using Approximate Bayesian Computation.

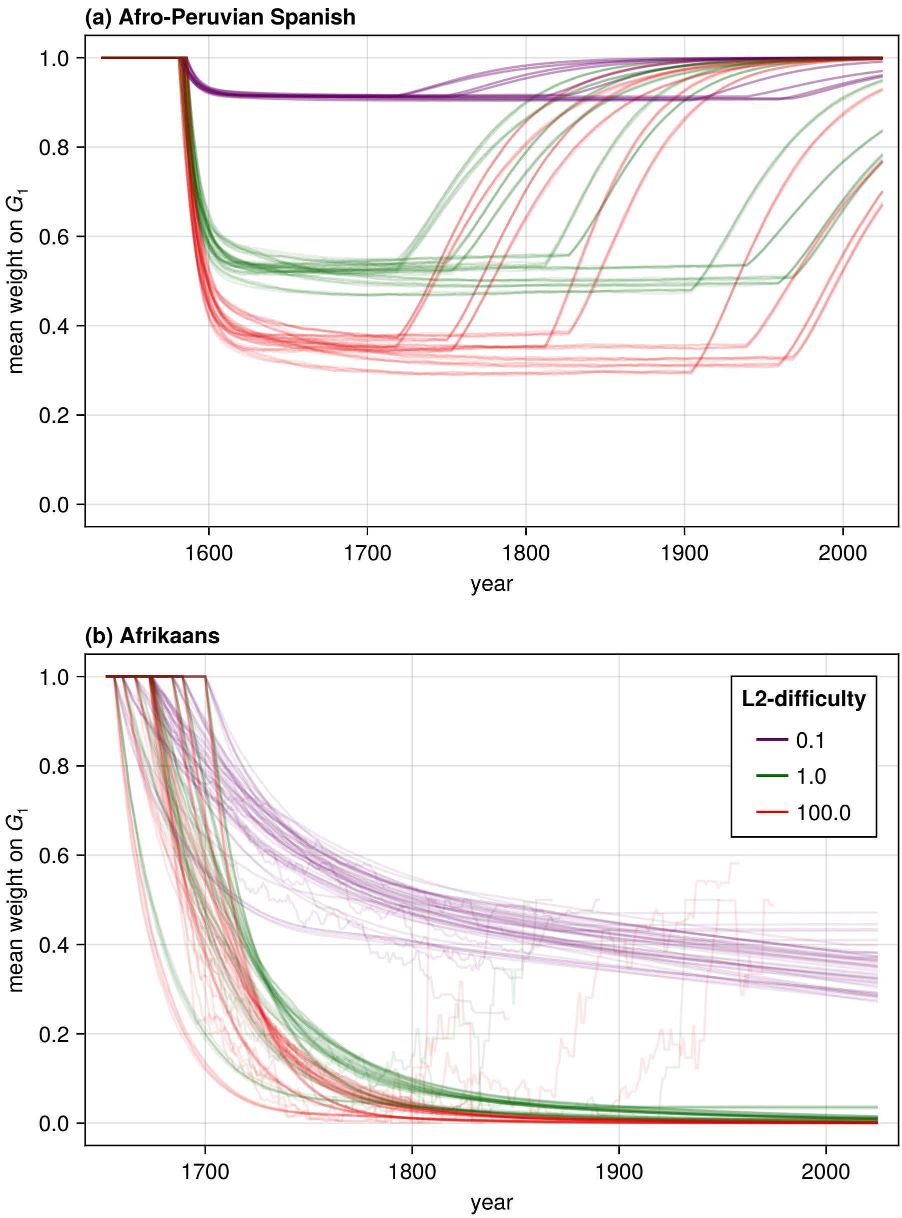

Figure 12: Simulations of Afrikaans and Afro-Peruvian Spanish with simulation parameters sampled from the posterior distributions of the sociodynamic parameters found using Approximate Bayesian Computation (Figure 10 and Figure 11).

8 Conclusion

This chapter has taken up the mathematical model of contact-induced change proposed in Kauhanen (2022) and implemented the model in a series of fairly realistic agent-based simulations. The simulations involved finite (rather than infinite) populations, local (as opposed to well-mixing) interaction patterns and finite (as opposed to eternal) lifetimes for individual speakers. The central result of Kauhanen (2022) – the existence and location of a bifurcation threshold involving the proportion of L2 learners – was confirmed in these finite simulations. In addition to systematic sweeps over the model's parameter space, the two case studies of Kauhanen (2022) were revisited in detail, with the population-dynamic model parameters optimized with respect to historical demographic data.

These simulations have, however, only scratched the surface. In the current implementation, the model involves 18 parameters which are in principle free to vary independently of each other. Although some interesting parameter interactions were mentioned above – such as between lattice aspect ratio and L2-difficulty or the length of the immigration period – these interactions for the most part await further, more systematic study. On the other hand, it would be possible to break away from the current implementation, to explore other mechanisms of population dynamics. For instance, the model can readibly be implemented on arbitrary interaction networks rather than the simple lattice setup ~~here assumed~~. This opens up a line of investigation – the intersection of language dynamics and network science – which only a handful of studies have explored so far (Josserand et al. 2021, Ke et al. 2008, Fagyal et al. 2010, Kauhanen 2017) **REF**: probably need to add some blythe/croft stuff I'm now forgetting about

A Population dynamics

The purpose of this appendix is to prove that, given the simulation setup assumed above, the expected proportion of L2 learners in the population is completely determined by the birth and immigration rates β and τ:

$$\mathrm{E}[\sigma] = \frac{\tau}{\beta + \tau}.$$

More precisely, this is the asymptotic value of $\mathrm{E}[\sigma]$ if L2 learning is allowed to continue indefinitely. In practice, in the simulations conducted in this chapter, this number describes the peak proportion of L2 learners attained during the T_{L2} simulation steps when L2 learning occurs (cf. Figure 6).

Let X and Y refer to the numbers of L1 and L2 learners in the population, respectively, at some given simulation step. Moreover, let X' and Y' refer to these numbers at the immediately following simulation step. Let B refer to the number of L1 learners born at a given step, and let M refer to the number of L2 learners immigrated. With the assumptions made in the main text, we have $B \sim \mathrm{Bin}(\lfloor N(1-N/K) \rfloor^+, \beta)$, where $N = X + Y$ is the total population size, K is a constant (the carrying capacity), and $\lfloor a \rfloor^+ = \lfloor a \rfloor$ if $a > 0$ and $\lfloor a \rfloor^+ = 0$ otherwise. Similarly, $M \sim \mathrm{Bin}(\lfloor N(1-N/K) \rfloor^+, \tau)$.

The expected numbers of L1 and L2 learners after one iteration are thus

$$\begin{cases} \mathrm{E}[X'] = \mathrm{E}[X] + \mathrm{E}[B] - \mu\mathrm{E}[X] \\ \mathrm{E}[Y'] = \mathrm{E}[Y] + \mathrm{E}[M] - \mu\mathrm{E}[Y] \end{cases}$$

where μ is a death rate (the exact value of this parameter will depend on the parameterization of the Weibull distribution assumed in the main text). The expected change between two consecutive steps is thus

$$\begin{cases} \mathrm{E}[X' - X] = \mathrm{E}[B] - \mu\mathrm{E}[X] \\ \mathrm{E}[Y' - Y] = \mathrm{E}[M] - \mu\mathrm{E}[Y] \end{cases}$$

Given that B and M are binomially distributed, the following holds:

$$\begin{cases} \mathrm{E}[X' - X] = \beta\lfloor N(1-N/K) \rfloor^+ - \mu\mathrm{E}[X] \\ \mathrm{E}[Y' - Y] = \tau\lfloor N(1-N/K) \rfloor^+ - \mu\mathrm{E}[Y] \end{cases}$$

Approximately, we have

$$\begin{cases} \mathrm{E}[X' - X] \approx \beta N(1-N/K) - \mu\mathrm{E}[X] \\ \mathrm{E}[Y' - Y] \approx \tau N(1-N/K) - \mu\mathrm{E}[Y] \end{cases}$$

At equilibrium, one has $\mathrm{E}[Y' - Y] = 0$ and so

$$\tau N(1 - N/K) = \mu\mathrm{E}[Y].$$

Taking expectations on both sides and dividing τ onto the right hand side,

$$\mathrm{E}[N] - \mathrm{E}[N]^2/K = \frac{\mu}{\tau}\mathrm{E}[Y].$$

Dividing both sides by $\mathrm{E}[N]$ we obtain

$$1 - \mathrm{E}[N]/K = \frac{\mu}{\tau}\mathrm{E}[\sigma],$$

from which

$$\mathrm{E}[\sigma] = \frac{\tau}{\mu}(1 - \mathrm{E}[N]/K). \tag{.13}$$

Carrying out the same exercise starting from $\mathrm{E}[X' - X] = 0$, one obtains

$$\mathrm{E}[\sigma] = 1 - \frac{\beta}{\mu}(1 - \mathrm{E}[N]/K). \tag{.14}$$

Equating (.13) and (.14) allows us to solve for $\mathrm{E}[N] = \mathrm{E}[X+Y]$, the expected total population size at equilibrium:

$$\mathrm{E}[N] = K\left(1 - \frac{\mu}{\beta + \tau}\right).$$

Substituting this back to (.13) finally gives us

$$\mathrm{E}[\sigma] = \frac{\tau}{\mu}\left(1 - \left(1 - \frac{\mu}{\beta+\tau}\right)\right) = \frac{\tau}{\mu} \cdot \frac{\mu}{\beta+\tau} = \frac{\tau}{\beta+\tau}$$

as desired.

Thus, in order to obtain a desired L2 learner proportion σ given a fixed birth rate β, we need to select the immigration rate τ such that

$$\tau = \frac{\sigma}{1-\sigma}\beta.$$

B Parameter estimation

To estimate population-dynamical parameters for Afro-Peruvian Spanish and Afrikaans, Approximate Bayesian Computation Sequential Monte Carlo (ABC-SMC) with Differential Evolution (DE) moves was used (**REF**S), as implemented in the ABCdeZ.jl package of the Julia programming language. The distance function between model prediction and empirical data is given by

$$\rho(\vec{\theta}) = \frac{1}{|T|}\sum_{t\in T}\left| O_{\mathrm{L1}}(t) - P_{\mathrm{L1}}(t,\vec{\theta})\right| + \frac{1}{|T|}\sum_{t\in T}\left| O_{\mathrm{L2}}(t) - P_{\mathrm{L2}}(t,\vec{\theta})\right|,$$

i.e. by the mean absolute error (this has the straightforward interpretation of being the average per-year error in total population size between model and data). Here, T denotes the set of time points in the data (i.e. the years for which demographic data is available), $O_{\text{L1}}(t)$ denotes the empirical (observed) number of L1 speakers at time t, $P_{\text{L1}}(t, \vec{\theta})$ denotes the modelled (predicted) number of L1 speakers at time t, given parameter vector $\vec{\theta}$, and so on. In respect ~~of~~ the empirical demographic data, I take all Europeans to be L1 speakers and all non-Europeans to be L2 speakers (at least initially, i.e. at the time they are introduced into the population). This constitutes an idealization, of course, but in the absence of further empirical information it is difficult to see how one might proceed otherwise.

The prior distributions assumed for the model parameters were either discrete or continuous uniform depending on the nature of the parameter; the lower and upper bounds of these prior distributions are listed in Table 10. The procedure was continued until either $\rho(\vec{\theta})$ stood below 500 or 100,000 simulations had been conducted. For Afro-Peruvian Spanish, the desired level of distance ($\rho(\vec{\theta}) = 500$) was attained. In the case of Afrikaans, $\rho(\vec{\theta})$ eventually stabilized at 1347.2 and did not improve from there.

Table 10: The prior distribution for each parameter was assumed to be uniform, with lower and upper bounds as described here.

Parameter	Symbol	Lower	Upper
Birth rate	β	0.001	0.1
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$	0.01	0.99
Length of immigration period	T_{L2}	10.0	1000.0
Mean L2 learner age at immigration	m	10.0	50.0
S.D. of L2 learner age at immigration	s	1.0	100.0
Number of L1 speakers at time 0	N_0	10.0	1000.0
Shape parameter of death distribution	k	1.0	10.0
Scale parameter of death distribution	λ	10.0	75.0
Population carrying capacity	K	10000.0	500000.0
Language contact start time	T^0_{L2}	1.0	100.0

References

Bezanson, Jeff, Alan Edelman, Stefan Karpinski & Viral B Shah. 2017. Julia: a fresh approach to numerical computing. *SIAM Review* 59(1). 65–98.

Bini, Milena. 1993. La adquisición del italiano: más allá de las propiedades sintácticas del parámetro pro-drop. In Juana M. Liceras (ed.), *La lingüística y el análisis de los sistemas no nativos*, 126–139. Ottawa: Dovehouse Editions Canada.

Bowser, Frederick P. 1974. *The African slave in Colonial Peru 1524–1650*. Stanford, CA: Stanford University Press.

Bush, Robert R. & Frederick Mosteller. 1955. *Stochastic models for learning*. New York, NY: Wiley.

Das, Swagatam & Ponnuthurai Nagaratnam Suganthan. 2011. Differential evolution: a survey of the state-of-the-art. *IEEE Transactions on Evolutionary Computation* 15(1). 4–31.

Fagyal, Zsuzsanna, Samarth Swarup, Anna María Escobar, Les Gasser & Kiran Lakkaraju. 2010. Centers and peripheries: network roles in language change. *Lingua* 120. 2061–2079.

Geritz, Stefan A. H. & Éva Kisdi. 2012. Mathematical ecology: why mechanistic models? *Journal of Mathematical Biology* 65. 1411–1415. DOI: 10.1007/s00285-011-0496-3.

Gibson, Edward & Kenneth Wexler. 1994. Triggers. *Linguistic Inquiry* 25(3). 407–454.

Gilbert, Nigel. 2020. *Agent-based models*. Second edition. London: SAGE.

Giliomee, Hermann & Richard Elphick. 1979. The structure of European domination at the Cape, 1652–1820. In Richard Elphick & Hermann Giliomee (eds.), *The shaping of South African society, 1652–1820*, 359–390. Cape Town: Longman.

Guerrero Montero, Juan, Andres Karjus, Kenny Smith & Richard A. Blythe. 2025. Reliable detection and quantification of selective forces in language change. *Corpus Linguistics and Linguistic Theory* 21(1). 31–73. DOI: 10.1515/cllt-2023-0064.

Heycock, Caroline & Joel Wallenberg. 2013. How variational acquisition drives syntactic change: the loss of verb movement in Scandinavian. *Journal of Comparative Germanic Linguistics* 16. 127–157.

Josserand, Mathilde, Marc Allassonnière-Tang, François Pellegrino & Dan Dediu. 2021. Interindividual variation refuses to go away: a Bayesian computer model of language change in communicative networks. *Frontiers in Psychology* 12. 626118. DOI: 10.3389/fpsyg.2021.626118.

Kauhanen, Henri. 2017. Neutral change. *Journal of Linguistics* 53(2). 327–358. DOI: 10.1017/S0022226716000141.

Kauhanen, Henri. 2022. A bifurcation threshold for contact-induced language change. *Glossa: a journal of general linguistics* 7(1). 1–32. DOI: 10.16995/glossa.8211.

Ke, Jinyun, Tao Gong & William S.-Y. Wang. 2008. Language change and social networks. *Communications in Computational Physics* 3(4). 935–949.

Margaza, Panagiota & Aurora Bel. 2006. Null subjects at the syntax–pragmatics interface: evidence from Spanish interlanguage of Greek speakers. In Mary Grantham O'Brien, Christine Shea & John Archibald (eds.), *Proceedings of the 8th Generative Approaches to Second Language Acquisition Conference (GASLA 2006)*, 88–97. Somerville, MA: Cascadilla Proceedings Project.

Muysken, Pieter. 2013. Scenarios for language contact. In Raymond Hickey (ed.), *The handbook of language contact*, 265–281. Malden, MA: Wiley-Blackwell.

Narendra, Kumpati S. & Mandayam A. L. Thathachar. 1989. *Learning automata: an introduction*. Englewood Hills, NJ: Prentice-Hall.

Niyogi, Partha & Robert C. Berwick. 1996. A language learning model for finite parameter spaces. *Cognition* 61. 161–193.

Pérez-Leroux, Ana T. & William R. Glass. 1999. Null anaphora in Spanish second language acquisition: probabilistic versus generative approaches. *Second Language Research* 15(2). 220–249.

Ponelis, Fritz. 1993. *The development of Afrikaans*. Frankfurt am Main: Peter Lang.

Samaniego, Francisco J. 2014. *Stochastic modeling and mathematical statistics: a text for statisticians and quantitative scientists*. Boca Raton, FL: CRC Press.

Sessarego, Sandro. 2014. Afro-Peruvian Spanish in the context of Spanish creole genesis. *Spanish in Context* 11(3). 381–401.

Sessarego, Sandro. 2015. *Afro-Peruvian Spanish: Spanish slavery and the legacy of Spanish creoles*. Amsterdam: Benjamins.

Simonenko, Alexandra, Benoit Crabbé & Sophie Prévost. 2019. Agreement syncretization and the loss of null subjects: quantificational models for Medieval French. *Language Variation and Change* 31(3). 275–301. DOI: 10.1017/S0954394519000188.

Smaldino, Paul E. 2023. *Modeling social behavior: mathematical and agent-based models of social dynamics and cultural evolution*. Princeton, NJ: Princeton University Press.

Storn, Rainer & Kenneth Price. 1997. Differential Evolution – a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization* 11. 341–359. DOI: 10.1023/A:1008202821328.

Trudgill, Peter. 2011. *Sociolinguistic typology: social determinants of linguistic complexity*. Oxford: Oxford University Press.

Wallenberg, Joel, Henri Kauhanen, George Walkden & Caroline Heycock. 2025. Grammar competition and variational learning. In Adam Ledgeway, Edith Aldridge, Anne Breitbarth, Katalin É. Kiss, Joseph Salmons & Alexandra Simonenko (eds.), *The Wiley Blackwell companion to diachronic linguistics*. Wiley.

Yang, Charles D. 2000. Internal and external forces in language change. *Language Variation and Change* 12. 231–250. DOI: 10.1017/S0954394500123014.
Yang, Charles D. 2002. *Knowledge and learning in natural language*. Oxford: Oxford University Press.